

The Constants of Nature

$E_{\rm h}:\text{K}=\frac{\text{kelvin}}{\text{K}}\left(\frac{T_{\rm p}}{m_{\rm p}}\right)\left(1-14\,C_{\rm D}^{-4}\,\mathscr{K}_{\rm \Phi}\right)\boxtimes$	$m_{\rm de}=\frac{3s}{\mathscr{K}_3\mathscr{K}_4}\left(\frac{16}{4\pi}\right)\left(m_{\rm n}+m_{\rm +}\right)\left(1+\frac{Im(\omega_{\rm i})}{\mathscr{K}_2}\left(\frac{L_2}{14}\right)\mathscr{K}_{\rm r}^3\boxtimes\right)$	$\mathcal{E}_{\rm p}=\frac{\text{joule}}{\text{kg}}\left(\frac{t_{\rm p}^2\,q_{\rm p}}{t_{\rm p}^2}\right)\left(1-\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\mu_{\rm hc}=\frac{\mathscr{K}_1}{hc}\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}}{t_{\rm p}}\frac{q_{\rm p}}{m_{\rm e}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$c=\left(\frac{L_2}{T_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$F_{90}=\text{farad}\left(1+\frac{C_d}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\boxtimes\right)$	$K_{\rm j}=\frac{\mathscr{K}_1}{\pi}\left(\frac{t_{\rm p}}{l_{\rm p}}\frac{q_{\rm p}}{m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\boxtimes\right)$	$g_{\rm de}=\frac{2\pi}{8\text{K}}\left(1+\frac{16}{l_{\rm p}}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_2+\mathscr{K}_4\right)^2\boxtimes\right)$
$\text{K}:\text{E}_{\rm h}=\frac{\text{kelvin}}{\mathscr{K}_1^4}\left(\frac{m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+14\,C_{\rm D}^{-4}\,\mathscr{K}_{\Phi}\right)\boxtimes$	$m_{\rm u}=\frac{1}{\mathscr{K}_3\mathscr{K}_4}\sqrt{\frac{16}{4\pi}}\left(m_{\rm n}+m_{\rm +}\right)\left(1+\left(\frac{16}{2\pi}\right)\mathscr{K}_{\rm r}^3\boxtimes\right)$	$R_{\rm K}=\frac{2\pi}{\mathscr{K}_1^2}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm N}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}}{t_{\rm p}}\frac{q_{\rm p}}{m_{\rm e}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{1}{\text{m}}:\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$H_{00}=\text{henry}\left(1-\frac{C_d}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\boxtimes\right)$	$e=\frac{\mathscr{K}_1}{h}\left(\frac{t_{\rm p}}{l_{\rm p}}\frac{q_{\rm p}}{m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\boxtimes\right)$	$\frac{\mu_{\rm de}}{\mu_{\rm N}}=\frac{2\pi}{8\text{K}}\left(1+\frac{16}{l_{\rm p}}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_2+\mathscr{K}_4\right)^2\boxtimes\right)$
$\Delta_{\rm VC1}=\frac{E_1}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2}\frac{m_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Phi}^2\boxtimes\right)$	$m_{\rm +}=\frac{-2\pi\mathscr{K}_1}{\mathscr{K}_3\mathscr{K}_4}\left(\frac{\text{GeV}}{\sqrt{5}}\right)\left(\frac{t_{\rm p}^2}{l_{\rm p}^2}\right)\left(1+\frac{1}{6^{1/2}}\left(\frac{2\pi}{2s}\right)\mathscr{K}_{\rm r}^3\boxtimes\right)$	$Z_0=4\pi\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$r_{\rm e}=\mathscr{K}_1^{-2}\left(\frac{t_{\rm p}}{m_{\rm e}}\right)\left(1-\sqrt{6\,C_{\rm R1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$\Omega_{00}=\text{ohm}\left(1-\frac{C_d}{6}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4\right)\boxtimes\right)$	$\Phi_0=\frac{\pi}{\mathscr{K}_1}\frac{t_{\rm p}^2}{2}\frac{m_{\rm p}}{m_{\rm p}}\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\boxtimes\right)$	$\frac{\mu_{\rm de}}{\mu_{\rm B}}=\frac{2\pi}{8\text{K}}\left(\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+\frac{16}{l_{\rm p}}\left(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_2+\mathscr{K}_4\right)^2\boxtimes\right)$
$\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2}\frac{m_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Phi}^2\boxtimes\right)$	$m_{\rm he}=\frac{\sqrt{3}}{\mathscr{K}_1^2\mathscr{K}_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_{\rm p}^2}{l_{\rm p}^2}\right)\left(1-\frac{3^{-1}}{\mathscr{K}_1\mathscr{K}_2}\left(\frac{s}{18}\right)\mathscr{K}_{\rm r}^3\boxtimes\right)$	$Z_{\rm p}=\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}}}\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$a_0=\frac{1}{\mathscr{K}_1^2}\left(\frac{t_{\rm p}}{m_{\rm e}}\right)\left(1-\sqrt{6\,C_{\rm R1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$ST_0=\frac{5}{2}+\log\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{l_{\rm p}^2\,T_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{P_{\rm p}}\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\mathscr{K}_2}\right)\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4\right)\boxtimes\right)$	$\frac{h}{e}=\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\frac{m_{\rm p}}{m_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\boxtimes\right)$	$b^{\prime}=\frac{v_{\rm peak}}{2\pi}\left(\frac{1}{t_{\rm p}\,T_{\rm p}}\right)\left(1+\frac{16}{16}\left(\frac{2\pi}{s}\right)^2\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_2-\mathscr{K}_4\right)^2\boxtimes\right)$
$\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_{\rm p}}{l_{\rm p}^2}\frac{m_{\rm p}}{m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Phi}^2\boxtimes\right)$	$\frac{G}{hc}=\text{GeV}^2\left(\frac{t_{\rm p}^2}{l_{\rm p}^2m_{\rm p}}\right)^2\left(1-\left(\frac{4\pi}{18}\right)\mathscr{K}_{\rm r}^3\boxtimes\right)$	$\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_{\rm p}}{l_{\rm p}}\right)\left(1+\frac{1}{2}Im\left(i^{i^{\prime\prime}}\right)\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm N}^{'}}{S}=\frac{2\,\mathscr{K}_1\mathscr{K}_2^2}{S}\left(1-18\left(6\pi\right)\mathscr{K}_1\mathscr{K}_3\mathscr{K}_4\boxtimes\right)$	$\sigma_{\rm p}=\sqrt{\frac{32}{18}}\,2\pi\mathscr{K}_1^4\left(\frac{t_{\rm p}}{l_{\rm p}}\frac{m_{\rm p}}{m_{\rm e}}\right)^2\left(1-2\sqrt{6\,C_{\rm R1}}\left(\mathscr{K}_1-\mathscr{K}_2\right)\boxtimes\right)$	$ST_1=\frac{5}{2}+\log\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{l_{\rm p}^2\,T_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{T_{\rm p}\,P_{\rm p}}\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\mathscr{K}_2}\right)\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4\right)\boxtimes\right)$	$c_1=(2\pi)^2\left(\frac{t_{\rm p}^4\,m_{\rm p}}{t_{\rm p}^3}\right)\left(1-\frac{4\pi^2}{32}\left(\mathscr{K}_1+\mathscr{K}_2\right)^2\boxtimes\right)$	$\mathcal{K}:\text{Hz}=\frac{\text{kelvin}}{2\pi}\left(\frac{1}{t_{\rm p}\,T_{\rm p}}\right)\left(1+\frac{16}{16}\left(\frac{2\pi}{s}\right)^2\left(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_2-\mathscr{K}_4\right)^2\boxtimes\right)$
$\text{Hz}:\text{eV}=\frac{2\pi\text{joule}}{\text{second eV}}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\right)\left(1+\sqrt{\zeta(2)}\,\$							

$t_p = 5.39125836832313 \times 10^{-40}$ s	Planck time
$l_p = 1.61625918175645 \times 10^{-38}$ m	Planck length
$q_p = 1.87554596713962 \times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795 \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \times 10^{-8}$ kg	Planck mass
$\Xi = 0.999999199973626 \times 10^{-7}$	hyperbolic universe boundary
5	number of dimensions
7	break in scale symmetry
$\pi_1 = 0.0854254531533304 \dots$	1 st hyperbolic partition constant
$\pi_2 = 3.66756753480501 \dots$	2 nd hyperbolic partition constant
$\pi_3 = 1.867496903900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\pi_4 = 1.87864963900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\pi_5 = 4.47826244916751 \dots$	hyperbolic radius constant
$\pi_6 = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{00} = 1.01494160640965 \dots$	Gieseking's constant
$V_8 = 0.02598321281930 \dots$	figure eight knot hyperbolic volume
$K = 0.9159655541772190 \dots$	Catalan's constant
$C_d = 0.2916506904030818 \dots$	domino tiling constant
$D_4 = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.7649747018528596 \dots + 1.3249706214087 \dots i$	omega_1 constant
$\omega_2 = 1.529594037057192 \dots$	omega_2 constant
$i^{i^{i^i}}$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\phi_i = (-1)^{1/3}$	imaginary golden ratio
$G_9 = 1.15172847301812 \dots$	tether length for grazing half the unit circle
$\pi = 3.14159265358979 \dots$	Archimedes' constant
$S = 2.29558714939263 \dots$	universal parabolic constant
$s = 5.244115110858423 \dots$	arc length of the unit lemniscate
$L = 2.62205755442919 \dots$	lemniscate constant
$L_1 = 1.31102877146201 \dots$	1 st lemniscate constant
$L_2 = 0.599070117367796 \dots$	2 nd lemniscate constant
$C_{ij} = 0.884713084793979 \dots$	ubiquitous constant
$G_{66} = 0.834626841674073 \dots$	Gauss's constant
$C_{CF} = 0.697774657964819 \dots$	continued fraction constant
$R_{FR} = 0.9981360445908509 \dots$	Ramanujan's 1 st continued fraction constant
$F = 1.226742107202305 \dots$	Fibonacci factorial constant
$M = -1.74756459463318 \dots$	Modeling constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.822825249678847 \dots$	Sierpinski's constant
$C_{JF} = 2.404825569575772 \dots$	1 st root of the Bessel function
$\alpha_{\text{beta}} = 1.19967864025773 \dots$	Real fixed point of the hyperbolic cotangent
$D_{90} = 0.739085133215160 \dots$	Dottie number
$L_1 = 0.662743419349181 \dots$	Laplace limit
$\alpha_F = 2.50290787509589 \dots$	alpha Feigenbaum constant
$\delta = 4.66920160910299 \dots$	delta Feigenbaum constant
$S = 3.24697960371714 \dots$	silver constant
$p = 1.3247179574474 \dots$	plastic constant
$C_Q = 0.605443657196732 \dots$	QRS constant
$\alpha_{\text{QRS}} = 2.03816937970215 \dots$	associated QRS constant
$\chi_{\text{MSM}} = 0.3532363781854995 \dots$	Hafner-Sarnak-McCurley constant
$T_0 = 2.731500000000000 \dots \times 10^5$ pascal	freezing point of water
$p_1 = 1.000000000000000 \dots \times 10^5$ pascal	1 bar
$E_1 = 1.01325003754773 \dots \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \dots \times 10^{24}$ joules	energy associated with Δv_{CS}
$f_1 = 5.400000000000000 \dots \times 10^{14}$ Hz	frequency of max light wavelength
$\lambda_{\text{peak}} = 4.96511423174427 \dots$	peak λ of spectral radiance/unit wavelength
$\nu_{\text{peak}} = 2.82143937212207 \dots$	peak emission of spectral flux/unit frequency
$\log(x) =$ natural logarithm	
$\Gamma(x) =$ gamma function	ln = lumen = cd · sr
$\zeta(x) =$ Riemann zeta function	N = newton = m kg/s ²
$\{n\} =$ derangement function	N = noether = m kg/s
	$\rho = \text{ohm} = \text{m}^2 \text{ kg/s}^2 \text{ C}^2$
units	$\text{Pa} = \text{pascal} = \text{kg/m}^2 \text{ s}^2$
$m =$ meter	$S = \text{siemens} = \text{S} = \text{C}^2/\text{m}^2 \text{ kg}$
$m =$ second	$T = \text{tesla} = \text{kg/C}^2$
$C =$ coulomb	$V = \text{volt} = \text{m}^2 \text{ kg/s}^2 \text{ C}$
$K =$ kelvin	$W = \text{watt} = \text{m}^2 \text{ kg/s}^3$
$\text{kg} =$ kilogram	$\text{Wh} = \text{weber} = \text{m}^2 \text{ kg/s} \cdot \text{C}$
derived units	
MHz = megahertz = 1.000000000000000 $\dots \times 10^6$ 1/s	
fm = femtometer = 1.000000000000000 $\dots \times 10^{-15}$ m	
eV = electron-volt = 1.000000000000000 $\dots$ eV	
MeV = megaelectron-volt = 1.000000000000000 $\dots \times 10^6$ eV	
GeV = gigaelectron-volt = 1.000000000000000 $\dots \times 10^9$ eV	
$r_n = \frac{4\pi}{(\sqrt{5})^n}$	$\frac{r_n}{r_k} = \frac{r_n}{r_k}$
$\frac{r_n}{r_k} = \frac{r_n}{r_k}$	$\frac{r_n}{r_k} = \sqrt{\frac{C_U}{4}}$
$\frac{r_n}{r_k} = \frac{r_n}{r_k}$	$\frac{r_n}{r_{de}} = \sqrt{\frac{2\pi}{\log(36)}}$
$\frac{r_n}{r_k} = \frac{r_n}{r_k}$	$E_p = \frac{1}{2} \frac{m_p}{r_p}$
external transform space	Physics Monastery
$V_{24} = \left(\frac{4\pi}{5!} \right) \left(\frac{4\pi}{15} \right) \left(\frac{4\pi}{35} \right) \left(\frac{4\pi}{18} \right) \left(\frac{4\pi}{32} \right) \left(\frac{4\pi}{8} \right$	